

B.4.2

Interface list

Interface list

SUK400LNII / LSI100901

Project-No.:	P24-152
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Revision history			
Date	Version	Comment	Author
10.01.2025	a	Als built FAT	Cirap

...	Module	Rack	Slot	I address	Q address	Type	Article no.	Firmware
	▼ ABX-PRT	0	0			RT Standard	Anybus X-gateway	V3.29
	► Interface	0	0 X1			ABX-PRT		
	Control	0	1	400...401		Input 002 bytes		
	Value	0	2	500...531		Input 032 bytes		
	States	0	3		400...403	Output 004 bytes		
	Value 1	0	4		500...563	Output 064 bytes		
	Value 2	0	5		564...595	Output 032 bytes		
	Alarms	0	6		700...715	Output 016 bytes		
		0	7					
		0	8					
		0	9					

On customer DCS Profinet configuration:
LAUDA INPUTS need to be
OUTPUTS
LAUDA OUTPUTS need to be
INPUTS

Customer DCS Profinet
configuration:

Steckplatz: 0

Kataloginformation

Kurzbezeichnung: RT Standard

Beschreibung: Dieser Device Access Point unterstützt RT-kommunikation

Artikel-Nr.: Anybus X-gateway

Firmware-Version: V3.29

Hardware-Erzeugnisstand: 2

GSD-Datei: gsdml-v2.3-hms-anybus_x_gateway_profinet_io-20161110.xml

Ausgabestand ändern

Modulbeschreibung aktualisieren

Profinet gateway

HMS Anybus x gateway

AB7651-F

Link HMS homepage

Link HMS homepage

<https://www.hms-networks.com/p/ab7651-f-anybus-x-gateway-profinet-io-device-profinet-io-device>

Mapping Profinet gateway

From gatew From LAUDA to DCS

From profir From DCS to LAUDA

Signal exchange DCS to LAUDA PLC

DB1000

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
	Control	Struct								
1	Watchdog	Bool	0.0	Watch dog	n/a	n/a	n/a			
2	Start/Stop	Bool	0.1	Start=1 , Stop =0	n/a	n/a	n/a			
3	Reset	Bool	0.2	Reset = 1	n/a	n/a	n/a			
4	Remote	Bool	0.3	Control Remote = 1	n/a	n/a	n/a			
5	0.4	Bool	0.4	spare	n/a	n/a	n/a			
6	0.5	Bool	0.5	spare	n/a	n/a	n/a			
7	0.6	Bool	0.6	spare	n/a	n/a	n/a			
8	0.7	Bool	0.7	spare	n/a	n/a	n/a			
9	1.0	Bool	1.0	spare	n/a	n/a	n/a			
10	1.1	Bool	1.1	spare	n/a	n/a	n/a			
11	1.2	Bool	1.2	Spare	n/a	n/a	n/a			
12	1.3	Bool	1.3	Spare	n/a	n/a	n/a			
13	1.4	Bool	1.4	Spare	n/a	n/a	n/a			
14	1.5	Bool	1.5	Spare	n/a	n/a	n/a			
15	1.6	Bool	1.6	Spare	n/a	n/a	n/a			
16	1.7	Bool	1.7	Spare	n/a	n/a	n/a			
	Value	Struct								
17	Temp Setpoint N10011	Real	2.0	Setpoint Temperature outflow	°C	-50	50			
18	6.0	Rela	6.0	spare	n/a	n/a	n/a			
19	10.0	Real	10.0	spare	n/a	n/a	n/a			
20	14.0	Real	14.0	spare	n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
21	Speed Pump N14011	Real	18.0	Speed Frequency converter Pump	%	60	100.0			
22	22.0	Real	22.0	spare	n/a	n/a	n/a			
23	26.0	Real	26.0	spare	n/a	n/a	n/a			
24	30.0	Real	30.0	spare	n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign

Signal exchange LAUDA PLC to DCS

DB1001

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
	Demand bool									
1	Watchdog	Bool	0.0	Watchdog	n/a	n/a	n/a			
2	Operation	Bool	0.1	Pump Operation	n/a	n/a	n/a			
3	V50011 On	Bool	0.2	Compressor V50011 On	n/a	n/a	n/a			
4	V60011 On	Bool	0.3	Compressor V60011 On	n/a	n/a	n/a			
5	Remote	Bool	0.4	0= Local ,1=Remote	n/a	n/a	n/a			
6	Filling	Bool	0.5	Filling	n/a	n/a	n/a			
7	Heating	Bool	0.6	Heating	n/a	n/a	n/a			
8	Cooling	Bool	0.7	Cooling	n/a	n/a	n/a			
9	Ready	Bool	1.0	Chiller ready	n/a	n/a	n/a			
10	Common fault	Bool	1.1	Common fault	n/a	n/a	n/a			
11	Common warning	Bool	1.2	Common warning (TCU still running)	n/a	n/a	n/a			
12	Critical error	Bool	1.3	Critical error (TCU Stop)	n/a	n/a	n/a			
13	1.4	Bool	1.4		n/a	n/a	n/a			
14	1.5	Bool	1.5		n/a	n/a	n/a			
15	1.6	Bool	1.6		n/a	n/a	n/a			
16	1.7	Bool	1.7		n/a	n/a	n/a			
17	2.0	Bool	2.0		n/a	n/a	n/a			
18	2.1	Bool	2.1		n/a	n/a	n/a			
19	2.2	Bool	2.2		n/a	n/a	n/a			
20	2.3	Bool	2.3		n/a	n/a	n/a			
21	2.4	Bool	2.4		n/a	n/a	n/a			
22	2.5	Bool	2.5		n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
23	2.6	Bool	2.6		n/a	n/a	n/a			
24	2.7	Bool	2.7		n/a	n/a	n/a			
25	3.0	Bool	3.0		n/a	n/a	n/a			
26	3.1	Bool	3.1		n/a	n/a	n/a			
27	3.2	Bool	3.2		n/a	n/a	n/a			
28	3.3	Bool	3.3		n/a	n/a	n/a			
29	3.4	Bool	3.4		n/a	n/a	n/a			
30	3.5	Bool	3.5		n/a	n/a	n/a			
31	3.6	Bool	3.6		n/a	n/a	n/a			
32	3.7	Bool	3.7		n/a	n/a	n/a			
	Value									
33	Actual Setpoint	Real	4.0	Actual Setpoint TIC N10011	°C	-50	50			
34	B10011	Real	8.0	Temperature probe HTF outflow	°C	n/a	n/a			
35	B10511	Real	12.0	Temperature probe HTF return	°C	n/a	n/a			
36	B10512	Real	16.0	Temperature probe HTF return	°C	n/a	n/a			
37	B14011	Real	20.0	Temperature probe tank	°C	n/a	n/a			
38	B16411	Real	24.0	Pressure sensor HTF outflow	bar	-1.0	15.0			
39	B18011	Real	28.0	Level sensor filling	%	0.0	100.0			
40	B55111	Real	32.0	Temperature probe outflow evaporator	°C	n/a	n/a			
41	B55211	Real	36.0	Pressure sensor outflow evaporator	bar	0.0	32.0			
42	B55811	Real	40.0	Pressure sensor discharge oil pump	bar	0.0	32.0			
43	B56511	Real	44.0	Temperature probe pressure gas line	°C	n/a	n/a			
44	B56811	Real	48.0	Pressure sensor condenser inlet	bar	0.0	32.0			
45	B57311	Real	52.0	Temperature probe inlet injection valve	°C	n/a	n/a			
46	B65111	Real	56.0	Temperature probe outflow evaporator	°C	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
47	B65211	Real	60.0	Pressure sensor outflow evaporator	bar	0.0	32.0			
48	B65511	Real	64.0	Temperature probe suction side compressor	°C	n/a	n/a			
49	B65811	Real	68.0	Pressure sensor discharge oil pump	bar	0.0	32.0			
50	B66511	Real	72.0	Temperature probe pressure gas line	°C	n/a	n/a			
51	B66811	Real	76.0	Pressure sensor condenser inlet	bar	0.0	32.0			
52	B67311	Real	80.0	Temperature probe inlet injection valve	°C	n/a	n/a			
53	N10011	Real	84.0	Actuating signal temperature controller TIC N10011	%	0.0	100.0			
54	88.0	Real	88.0	spare	%	0.0	100.0			
55	spare	Real	92.0	spare	n/a	n/a	n/a			
56	spare	Real	96.0	spare	n/a	n/a	n/a			
	Alarms									
1	E-Stop internal	Bool	0.0	Emergency stop	n/a	n/a	n/a			
2	E-Stop external	Bool	0.1	Emergency stop external	n/a	n/a	n/a			
3	230VAC	Bool	0.2	F1 Fuse power supply 230V AC	n/a	n/a	n/a			
4	24VDC	Bool	0.3	F2 Power supply 24V DC	n/a	n/a	n/a			
5	24VDC_Puffer	Bool	0.4	G10 Power supply 24VDC buffer	n/a	n/a	n/a			
6	Profinet	Bool	0.5	Profinet common fault	n/a	n/a	n/a			
7	Communication DCS	Bool	0.6	Fault communication interface customer	n/a	n/a	n/a			
8	SF Safety SPS	Bool	0.7	System fault safety system	n/a	n/a	n/a			
9	F11	Bool	1.0	F11 Motor circuit breaker	n/a	n/a	n/a			
10	F12Y,F12YY	Bool	1.1	F12Y,F12YY Motor circuit breaker	n/a	n/a	n/a			
11	F13Y,F13YY	Bool	1.2	F13Y,F13YY Motor circuit breaker	n/a	n/a	n/a			
12	F15	Bool	1.3	F15 Motor circuit breaker	n/a	n/a	n/a			
13	F16	Bool	1.4	F16 Motor circuit breaker	n/a	n/a	n/a			
14	F10011	Bool	1.5	F10011 Flow limiter HTF	n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
15	F10511	Bool	1.6	F10511 Temperature limiter TZAH	n/a	n/a	n/a			
16	F10611	Bool	1.7	F10611 Temperature control TSAH	n/a	n/a	n/a			
17	F10811	Bool	2.0	F10811 Temperature control TSAL	n/a	n/a	n/a			
18	F15611	Bool	2.1	F15611 Thermistor control pump	n/a	n/a	n/a			
19	F18011	Bool	2.2	F18011 Level control LSAL	n/a	n/a	n/a			
20	F18111	Bool	2.3	F18111 Level monitoring LAL	n/a	n/a	n/a			
21	F18211	Bool	2.4	F18211 Level monitoring LAH	n/a	n/a	n/a			
22	F18311	Bool	2.5	F18311 Level control LSAH	n/a	n/a	n/a			
23	F50011	Bool	2.6	F50011 Temperature control compressor	n/a	n/a	n/a			
24	F50511	Bool	2.7	F50511 Pressure control oil	n/a	n/a	n/a			
25	F51011	Bool	3.0	F51011 High pressure cut-out (Reset)	n/a	n/a	n/a			
26	F51511	Bool	3.1	F51511 High pressure safety cut-out (Reset)	n/a	n/a	n/a			
27	F55611	Bool	3.2	F55611 Pressure control suction side compressor	n/a	n/a	n/a			
28	F56511	Bool	3.3	F56511 Temperature control pressure gas line	n/a	n/a	n/a			
29	F60011	Bool	3.4	F60011 Temperature control compressor	n/a	n/a	n/a			
30	F60511	Bool	3.5	F60511 Pressure control oil	n/a	n/a	n/a			
31	F61011	Bool	3.6	F61011 High pressure cut-out (Reset)	n/a	n/a	n/a			
32	F61511	Bool	3.7	F61511 High pressure safety cut-out (Reset)	n/a	n/a	n/a			
33	F65611	Bool	4.0	F65611 Pressure control suction side compressor	n/a	n/a	n/a			
34	F66511	Bool	4.1	F66511 Temperature control pressure gas line	n/a	n/a	n/a			
35	B10011	Bool	4.2	B10011 Sensor error	n/a	n/a	n/a			
36	B10511	Bool	4.3	B10511 Sensor error	n/a	n/a	n/a			
37	B14011	Bool	4.4	B14011 Sensor error	n/a	n/a	n/a			
38	B16411	Bool	4.5	B16411 Sensor error	n/a	n/a	n/a			
39	B16811	Bool	4.6	B16811 Sensor error	n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
40	B18011	Bool	4.7	B18011 Sensor error	n/a	n/a	n/a			
41	B55111	Bool	5.0	B55111 Sensor error	n/a	n/a	n/a			
42	B55211	Bool	5.1	B55211 Sensor error	n/a	n/a	n/a			
43	B55811	Bool	5.2	B55811 Sensor error	n/a	n/a	n/a			
44	B56511	Bool	5.3	B56511 Sensor error	n/a	n/a	n/a			
45	B56811	Bool	5.4	B56811 Sensor error	n/a	n/a	n/a			
46	B57311	Bool	5.5	B57311 Sensor error	n/a	n/a	n/a			
47	B65111	Bool	5.6	B65111 Sensor error	n/a	n/a	n/a			
48	B65211	Bool	5.7	B65211 Sensor error	n/a	n/a	n/a			
49	B65511	Bool	6.0	B65511 Sensor error	n/a	n/a	n/a			
50	B65811	Bool	6.1	B65811 Sensor error	n/a	n/a	n/a			
51	B66511	Bool	6.2	B66511 Sensor error	n/a	n/a	n/a			
52	B66811	Bool	6.3	B66811 Sensor error	n/a	n/a	n/a			
53	B67311	Bool	6.4	B67311 Sensor error	n/a	n/a	n/a			
54	H50011	Bool	6.5	H50011 Valve error	n/a	n/a	n/a			
55	H50511	Bool	6.6	H50511 Valve error	n/a	n/a	n/a			
56	H60011	Bool	6.7	H60011 Valve error	n/a	n/a	n/a			
57	H60511	Bool	7.0	H60511 Valve error	n/a	n/a	n/a			
58	H61011	Bool	7.1	H61011 Valve error	n/a	n/a	n/a			
59	N14011	Bool	7.2	N14011 Feedback signal missing	n/a	n/a	n/a			
60	K12Y	Bool	7.3	K12Y Feedback signal missing	n/a	n/a	n/a			
61	K12YY	Bool	7.4	K12YY Feedback signal missing	n/a	n/a	n/a			
62	K13Y	Bool	7.5	K13Y Feedback signal missing	n/a	n/a	n/a			
63	K13YY	Bool	7.6	K13YY Feedback signal missing	n/a	n/a	n/a			
64	K15	Bool	7.7	K15 Feedback signal missing	n/a	n/a	n/a			

No.	Name	Datatype	Offset	Comment	Unit	Range min	Range max	Test (Y/N)	Test Date	Sign
65	K16	Bool	8.0	K16 Feedback signal missing	n/a	n/a	n/a			
66	F16511	Bool	8.1	F16511 Pressure control HTF outflow PSAH	n/a	n/a	n/a			
67	VSD N14011	Bool	8.2	Frequency converter N14011	n/a	n/a	n/a			
68	8.3	Bool	8.3		n/a	n/a	n/a			
69	8.4	Bool	8.4		n/a	n/a	n/a			
70	8.5	Bool	8.5		n/a	n/a	n/a			
71	8.6	Bool	8.6		n/a	n/a	n/a			
72	8.7	Bool	8.7		n/a	n/a	n/a			
73	Byte 9	Byte	9.0	Spare	n/a	n/a	n/a			
74	Byte 10	Byte	10.0	Spare	n/a	n/a	n/a			
75	Byte 11	Byte	11.0	Spare	n/a	n/a	n/a			
76	Byte 12	Byte	12.0	Spare	n/a	n/a	n/a			
77	Byte 13	Byte	13.0	Spare	n/a	n/a	n/a			
78	Byte 14	Byte	14.0	Spare	n/a	n/a	n/a			
79	Byte 15	Byte	15.0	Spare	n/a	n/a	n/a			